

✓ AI-Powered Professional Email Generator

Project 1 — AI Automation Portfolio

This project uses Python and prompt engineering to generate professional emails based on user requirements.

✓ Step 2: Prompt Design

In this section, we define the information required to generate a professional email and create a structured prompt for the AI model.

```
# Email information provided by the user

email_purpose = "Request a meeting"
recipient = "Hiring Manager"
key_details = "I would like to discuss a potential internship opportunity next week."
tone = "Professional"
```

```
# Create a structured prompt for the AI model

prompt = f"""
You are a professional business communication assistant.

Write a clear and professional email using the following information:

Email purpose: {email_purpose}
Recipient: {recipient}
Key details: {key_details}
Tone: {tone}

Requirements:
1. Create a relevant email subject line.
2. Include a professional greeting.
3. Write a clear and concise email body.
4. Use the requested tone.
5. End with a polite closing.
6. Do not add explanations before or after the email.

Return only the completed email.
"""
```

```
print(prompt)
```

```
You are a professional business communication assistant.

Write a clear and professional email using the following information:

Email purpose: Request a meeting
Recipient: Hiring Manager
Key details: I would like to discuss a potential internship opportunity next week.
Tone: Professional

Requirements:
1. Create a relevant email subject line.
2. Include a professional greeting.
3. Write a clear and concise email body.
4. Use the requested tone.
5. End with a polite closing.
6. Do not add explanations before or after the email.

Return only the completed email.
```

✓ Step 2.2: Connect to the OpenAI API

In this section, we connect the structured prompt to an AI model and generate a professional email.

```
!pip install -q openai
```

```
from google.colab import userdata

api_key = userdata.get("OPENAI_API_KEY")
```

```
from openai import OpenAI

client = OpenAI(api_key=api_key)
```

```
client = OpenAI(api_key=api_key)
```

```
response = client.responses.create(
    model="gpt-5-mini",
    input=prompt
)

generated_email = response.output_text

print(generated_email)
```

Example Output after providing the exact OpenAI KEY

Subject: Request to Discuss Internship Opportunities

Dear Hiring Manager,

I hope you are doing well.

I am writing to express my interest in discussing potential internship opportunities with your organization. I would appreciate the opportunity to schedule a meeting next week to learn more about available roles and how my skills may contribute to your team.

Please let me know a convenient time for you.

Thank you for your time and consideration.

Best regards,

Bakht Zamin

✓ Test a New Email Scenario

Instead of rewriting the prompt each time, we will change only the input values like as following:

```
# Email information provided by the user

email_purpose = "Follow up after a job interview"
recipient = "Hiring Manager"
key_details = """
Thank the hiring manager for the interview.
Mention that I enjoyed learning about the role.
Reaffirm my interest in the position.
Ask if there are any updates regarding the next steps.
"""
tone = "Professional and enthusiastic"
```

```
# we will run this cell again
prompt = f"""
You are a professional business communication assistant.

Write a clear and professional email using the following information:

Email purpose: {email_purpose}
Recipient: {recipient}
Key details: {key_details}
Tone: {tone}

Requirements:
1. Create a relevant email subject line.
2. Include a professional greeting.
```

```

3. Write a clear and concise email body.
4. Use the requested tone.
5. End with a polite closing.
6. Do not add explanations before or after the email.

```

```

Return only the completed email.
"""

```

```

# Then
response = client.responses.create(
    model="gpt-5-mini",
    input=prompt
)

generated_email = response.output_text

print(generated_email)

```

✓ Improve the Prompt

Now we will make the prompt more specific and reliable.

We will replace the current prompt code with the following improved version:

```

prompt = f"""
You are an expert business communication assistant.

Your task is to write a polished, professional, and ready-to-send email.

Use the following information:

Email purpose: {email_purpose}
Recipient: {recipient}
Key details: {key_details}
Requested tone: {tone}

Follow these instructions carefully:

1. Create a concise and relevant subject line.
2. Begin with an appropriate professional greeting.
3. Clearly communicate the email's purpose.
4. Include all relevant information from the key details.
5. Use the requested tone consistently.
6. Keep the email concise and easy to read.
7. Avoid unnecessary repetition, exaggerated language, and generic filler.
8. End with a polite professional closing.
9. Add "Bakht Zamin" as a signature placeholder.
10. Return only the completed email.
11. Do not explain your writing choices.
12. Do not use markdown formatting.

Output format:

Subject: [Subject line]

[Greeting]

[Email body]

[Professional closing]
[Your Name]
"""

```

✓ Step 3.2: Interactive Email Generator

This section allows users to enter their email requirements and generate a professional email without manually editing the Python code.

```

# Collect email information from the user

email_purpose = input("What is the purpose of the email? ")
# Request a project deadline extension

```

```
recipient = input("Who is the recipient? ")
# Project Manager

key_details = input("What key details should be included? ")
# Request a three-day extension because additional time is needed to complete final testing and improve the project quality.

tone = input("What tone should the email use? ")
# Professional and respectful
```

Generate the Email Using the New Inputs

```
prompt = f"""
You are an expert business communication assistant.

Your task is to write a polished, professional, and ready-to-send email.

Use the following information:

Email purpose: {email_purpose}
Recipient: {recipient}
Key details: {key_details}
Requested tone: {tone}

Follow these instructions carefully:

1. Create a concise and relevant subject line.
2. Begin with an appropriate professional greeting.
3. Clearly communicate the email's purpose.
4. Include all relevant information from the key details.
5. Use the requested tone consistently.
6. Keep the email concise and easy to read.
7. Avoid unnecessary repetition, exaggerated language, and generic filler.
8. End with a polite professional closing.
9. Add "Bakht Zamin " as a signature placeholder.
10. Return only the completed email.
11. Do not explain your writing choices.
12. Do not use markdown formatting.

Output format:

Subject: [Subject line]

[Greeting]

[Email body]

[Professional closing]
[Bakht Zamin]
"""
```

```
# Run all the code cells
response = client.responses.create(
    model="gpt-5-mini",
    input=prompt
)

generated_email = response.output_text

print(generated_email)
```

